TASK 3: ELECTRICAL DIAGNOSTIC TESTING AND EVIDENCE RECORDING

Autocare Technician Level 2 | Block 6 | v2.1

TIME	EVIDENCE	PROGRAMME POINT
60 minutes	**Practical | Knowledge | Documentation**	**Weeks 51-60**

Your task

A vehicle has an electrical concern requiring structured testing. You must carry out safe electrical checks, record values and explain whether the evidence confirms the fault or requires further diagnosis.

What you must do

1. Confirm the reported electrical concern with the assessor/workplace mentor.
2. Prepare the vehicle/rig and select safe test points.
3. Carry out the electrical tests instructed by the assessor/workplace mentor.
4. Record results and compare with expected values.
5. Explain whether the evidence confirms the fault and reinstate the workstation.

What good evidence looks like

SAFE SETUP

Suitable PPE, a prepared work area and controlled vehicle or equipment.

CORRECT METHOD

A logical sequence using the relevant procedure and technical information.

ACCURATE EVIDENCE

Actual readings, findings and actions recorded clearly and precisely.

SOUND DECISIONS

Appropriate recommendations, referral or authorisation supported by evidence.

You will need

- Light vehicle or electrical training rig
- Multimeter and test leads
- Manufacturer wiring or technical information
- Battery support/PPE as required
- Apprentice Assessment Record

SOW ALIGNMENT

Block 6 - Systems Integration and Advanced Diagnostics
Duties: 1-10 KSB evidence focus: K12-K19, S8-S15, B1, B2
Task reference: Block_6_Task_3_Electrical_Diagnostic_Check

Apprentice, location and vehicle details

Apprentice name

Date

TASK COMPLETED AT

○ **APPRENTICE'S WORKPLACE**

○ **TRAINING CENTRE**

Vehicle make

Model

Registration

Mileage

Engine size / type

VIN / chassis number

Measurements and technical checks

Battery voltage		Test point voltage	
Voltage drop		Continuity result	
Resistance reading		Fuse/relay checked	
Relevant DTCs		Confirmed fault/evidence	

Special tools, equipment or calibration requirements

Task evidence

PPE and precautions used

Safety-critical findings and immediate action

Work completed, findings and recommendations

Knowledge check

1. Why should testing follow a logical sequence?

2. What safety precautions apply when using a multimeter?

Knowledge check - continued

3. (a) What can excessive voltage drop indicate? (b) Why is it important to record actual values rather than just pass/fail?

4. How would you avoid damaging an electrical circuit?

5. (a) When should a wiring diagram or technical data be used? (b) What further action is needed if the test does not confirm the fault?

Assessor/Workplace Mentor observation

- ☐ Use a structured electrical testing process.
- ☐ Carry out voltage, continuity or resistance checks safely.
- ☐ Record test results and compare with expected values.
- ☐ Explain findings using evidence rather than assumption.
- ☐ Complete diagnostic documentation accurately.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

○ **PASS** ○ **REFER**

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

PIN (if applicable)

Apprentice signature

Assessor/Workplace Mentor signature

Date

ADAS K17 — repair checks and calibration

For electrical diagnostic testing, use manufacturer information for a vehicle or a clearly labelled case study:
1. Explain whether the work could affect ADAS and identify the required checks.
2. Explain any relevant static and dynamic calibration methods and equipment, or why specialist advice is needed.
3. Record your information source and referral decision. Do not assume every repair requires calibration or carry out work beyond your authorised role.

Your answer

ADAS K18 — driver assistance and responsibilities

Use a clearly labelled driver-assistance electrical fault case. Identify the feature, whether it is passive or active, and how the described fault changes the warning or assistance available. Explain the driver’s responsibilities and, within the levels of driving automation, identify the level supported by the information provided. Name the information source; do not assume features or capabilities that the example does not show.

Your answer

BLOCK 6 | TASK 3

Autocare Level 2

Answer continuation

Use this page when an answer will not fit visibly in the original box. Reference the question and write "See continuation page" in the original answer. Keep all text visible for printing; add further numbered copies as needed.

Apprentice name

Additional page reference

Question / section reference

Continued answer

Question / section reference

Continued answer